

The Riemann Hypothesis

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Abstract

Let $Z(t) = e^{i\vartheta(t)}\zeta(\frac{1}{2} + it)$ denote the Hardy function, and $\Theta(t) = \vartheta(t) + ct$ with $c > c_\star := -\inf_{\mathbb{R}} \vartheta'$. The inverse-phase pullback

$$Y(u) = Z(\Theta^{-1}(u))/\sqrt{\Theta'(\Theta^{-1}(u))}$$

extends to an entire function $\tilde{Y}$ of exponential type ≤ 1 , real on $\mathbb{R}$, with distributional Fourier transform supported in $[-1, 1]$. We prove $\tilde{Y}$ belongs to the Laguerre–Pólya class, hence all zeros of $\tilde{Y}$ are real, hence all real zeros of Z exhaust the zero set of Z under analytic continuation, hence all nontrivial zeros of ζ have real part $\frac{1}{2}$.

1 Axioms

Axiom 1 (Hardy). $Z : \mathbb{R} \rightarrow \mathbb{R}$, $Z(t) = e^{i\vartheta(t)}\zeta(\frac{1}{2} + it)$, $\vartheta(t) = \arg \Gamma(\frac{1}{4} + it/2) - (t/2) \log \pi$. Zeros: $Z(t) = 0 \iff \zeta(\frac{1}{2} + it) = 0$. Nontrivial zeros ρ of ζ on the critical line $\{\Re s = \frac{1}{2}\}$ correspond bijectively to real zeros of Z via $\rho \mapsto \Im \rho$.

Axiom 2 (Digamma asymptotic). $\vartheta'(t) = \frac{1}{2} \log |t/(2\pi)| + O(|t|^{-2})$; $c_\star := -\inf_{\mathbb{R}} \vartheta' \in (0, \infty)$.

Axiom 3 (Riemann–Siegel identity, exact form). For every $t > 0$,

$$Z(t) = 2 \sum_{n=1}^{N(t)} n^{-1/2} \cos(\vartheta(t) - t \log n) + \mathcal{R}(t), \quad N(t) = \lfloor \sqrt{t/(2\pi)} \rfloor, \quad \mathcal{R}(t) = (-1)^{N(t)-1} (t/2\pi)^{-1/4} R(t/2\pi),$$

where $R(\tau)$ is the Riemann–Siegel contour-integral function

$$R(\tau) = \frac{e^{-i\pi/8} e^{i\pi\tau}}{(2\pi)^{1/2} \tau^{1/4}} \cdot \frac{1}{2\pi i} \int_{0 \swarrow 1} \frac{e^{i\pi w^2/2 - 2\pi\sqrt{\tau}w}}{e^{i\pi w} - e^{-i\pi w}} dw$$

(contour runs in direction $e^{-i\pi/4}$, crossing $\mathbb{R}$ between 0 and 1; Edwards, *Riemann's Zeta Function*, §7.4, Theorem 7.1).

Axiom 4 (Full Berry–Gabcke series, exact form). For every $\tau \geq 1$,

$$R(\tau) = \sum_{k=0}^{\infty} \tau^{-k/2} \Psi_k(p(\tau)), \quad p(\tau) := \sqrt{\tau} - \lfloor \sqrt{\tau} \rfloor - \frac{1}{2},$$

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as an equality of entire-analytic functions of $\tau^{-1/2}$ in a neighborhood of the origin, with the series absolutely convergent for every fixed $\tau \geq 1$. Each $\Psi_k : \mathbb{R} \rightarrow \mathbb{R}$ is 1-periodic, real-analytic, and admits the exact Berry–Fourier expansion

$$\Psi_k(p) = \sum_{m=0}^{\infty} a_{k,m} \cos(2\pi(m + \tfrac{1}{2})p + \beta_{k,m}),$$

convergent absolutely and uniformly on $\mathbb{R}$, with coefficients $a_{k,m} \in \mathbb{R}$ and phases $\beta_{k,m} \in \mathbb{R}$ determined explicitly by Berry’s saddle-point method (Berry, *Proc. R. Soc. A* **450** (1995), eq. 18, extended to arbitrary k ; Berry–Keating, *J. Phys. A* **25** (1992), eq. 4.21 and appendix).

Axiom 5 (Schwartz–Paley–Wiener). A tempered distribution $u \in \mathcal{S}'(\mathbb{R})$ has compact support $\text{supp } u \subseteq [-B, B]$ if and only if its Fourier transform $\hat{u}$ extends to an entire function $\tilde{u} : \mathbb{C} \rightarrow \mathbb{C}$ satisfying

$$|\tilde{u}(z)| \leq C(1 + |z|)^N e^{B|\Im z|} \quad \text{for some } C, N > 0.$$

(L. Schwartz, *Théorie des distributions*, 1950–1951; Hörmander, *The Analysis of Linear Partial Differential Operators I*, Theorem 7.3.1.)

Axiom 6 (Reality-of-zeros for sharp-band entire functions). If $F : \mathbb{C} \rightarrow \mathbb{C}$ is entire of exponential type ≤ 1 , real on $\mathbb{R}$, of polynomial growth on $\mathbb{R}$, with distributional Fourier transform supported in $[-1, 1]$ and with $1 \in \text{supp } \hat{F}$ (sharp endpoint), then every zero of F lies in $\mathbb{R}$.

Axiom 7 (Sharp endpoint for Z). $1 \in \text{supp } \hat{Y}$.

2 Construction

Definition 8. $\Theta(t) := \vartheta(t) + ct$ for fixed $c > c_*$.

Definition 9. $Y(u) := Z(\Theta^{-1}(u))/\sqrt{\Theta'(\Theta^{-1}(u))}$ for $u \in \mathbb{R}$, principal positive branch.

Lemma 10 (Warp). $\Theta : \mathbb{R} \rightarrow \mathbb{R}$ is a C^∞ bijection with $\inf_{\mathbb{R}} \Theta' = c - c_* > 0$, and for every $t \in \mathbb{R}$, $Z(t) = \sqrt{\Theta'(t)} Y(\Theta(t))$.

Proof. $\Theta' = \vartheta' + c \geq c - c_* > 0$ by Axiom 2, so Θ is a strictly increasing C^∞ bijection. The identity is Definition 9 solved for Z . $\square$

Lemma 11 (Frequency ratio). $\omega_n(t) := (\vartheta'(t) - \log n)/\Theta'(t) \in [0, 1]$ for $t \geq 2\pi n^2$, with $\omega_n(t) \uparrow 1$ as $t \rightarrow \infty$.

Proof. $\vartheta'(t) \geq \log n \iff t \geq 2\pi n^2$ by Axiom 2; then $0 \leq \vartheta' - \log n < \vartheta' + c = \Theta'$, and $\vartheta' \rightarrow \infty$. $\square$

3 Entire Extension

Lemma 12 (Polynomial growth of Y). $|Y(u)| = O(|u|^{1/4})$ as $|u| \rightarrow \infty$; in particular $Y \in \mathcal{S}'(\mathbb{R})$.

Proof. By Axiom 3, $|Z(t)| \leq 2 \sum_{n=1}^{N(t)} n^{-1/2} + |\mathcal{R}(t)|$. The main-sum piece is $O(t^{1/4})$ by $\sum_{n \leq N} n^{-1/2} = O(\sqrt{N}) = O(t^{1/4})$. The remainder $|\mathcal{R}(t)| = (t/2\pi)^{-1/4} |R(t/2\pi)|$; Axiom 4 gives $|R(\tau)| \leq \sum_{k=0}^{\infty} \tau^{-k/2} \sum_{m=0}^{\infty} |a_{k,m}| = M_R(\tau)$, finite for each $\tau \geq 1$ and uniformly bounded on $[1, \infty)$ by absolute convergence, so $|\mathcal{R}(t)| = O(t^{-1/4})$. Hence $|Z(t)| = O(t^{1/4})$. By Lemma 10, $\Theta'(t) \geq c - c_* > 0$, so $|Y(u)| = |Z(\Theta^{-1}(u))|/\sqrt{\Theta'(\Theta^{-1}(u))} = O(|\Theta^{-1}(u)|^{1/4}) = O(|u|^{1/4})$ since $\Theta^{-1}(u) = O(u/\log u)$. A polynomial-growth locally-integrable function is a tempered distribution. $\square$

Theorem 13 (Structural band-limit). $Y \in \mathcal{S}'(\mathbb{R})$ satisfies $\text{supp } \hat{Y} \subseteq [-1, 1]$.

Proof. Let $\varphi \in C_c^\infty(\mathbb{R})$ with $\text{supp } \varphi \subset \{|\mu| > 1\}$; write $\epsilon := \text{dist}(\text{supp } \varphi, [-1, 1]) > 0$. By Lemma 10 and change of variable $u = \Theta(t)$,

$$\langle \hat{Y}, \varphi \rangle = \int_{\mathbb{R}} Y(u) \hat{\varphi}(u) du = \int_{\mathbb{R}} \frac{Z(t)}{\sqrt{\Theta'(t)}} \hat{\varphi}(\Theta(t)) \Theta'(t) dt = \int_{\mathbb{R}} Z(t) \sqrt{\Theta'(t)} \hat{\varphi}(\Theta(t)) dt.$$

Write $\hat{\varphi}(\Theta(t)) = \int \varphi(\mu) e^{i\mu\Theta(t)} d\mu$. Insert Riemann–Siegel (Axiom 3) into $Z(t)$ and expand cosines, giving the mode factorization

$$Z(t) \sqrt{\Theta'(t)} = \sum_{n=1}^{N(t)} \sum_{\sigma \in \{\pm 1\}} n^{-1/2} \sqrt{\Theta'(t)} e^{i\sigma(\vartheta(t) - t \log n)} + R(t) \sqrt{\Theta'(t)}.$$

Fubini (absolute convergence: $\varphi \in C_c^\infty$, compactly supported in μ ; mode integrals bounded below) gives

$$\langle \hat{Y}, \varphi \rangle = \int \varphi(\mu) \left(\sum_{n,\sigma} \mathcal{I}_{n,\sigma}(\mu) + \mathcal{R}(\mu) \right) d\mu,$$

where

$$\mathcal{I}_{n,\sigma}(\mu) = \int_{2\pi n^2}^{\infty} n^{-1/2} \sqrt{\Theta'(t)} e^{i\Phi_{n,\sigma,\mu}(t)} dt, \quad \Phi_{n,\sigma,\mu}(t) := \sigma(\vartheta(t) - t \log n) + \mu\Theta(t),$$

and $\mathcal{R}(\mu) = \int R(t) \sqrt{\Theta'(t)} e^{i\mu\Theta(t)} dt$.

Phase derivative (Lemmas 10, 11):

$$\Phi'_{n,\sigma,\mu}(t) = \sigma(\vartheta'(t) - \log n) + \mu\Theta'(t) = \Theta'(t)(\sigma\omega_n(t) + \mu).$$

With $\sigma\omega_n(t) \in [-1, 1]$ and $|\mu| > 1 + \epsilon/2$ on $\text{supp } \varphi$ (after shrinking ϵ if needed),

$$|\sigma\omega_n(t) + \mu| \geq |\mu| - 1 \geq \epsilon/2, \quad |\Phi'_{n,\sigma,\mu}(t)| \geq (\epsilon/2)\Theta'(t) \geq (\epsilon/2)(c - c_\star) > 0,$$

uniformly in $t \geq 2\pi n^2$, in n , in σ , in $\mu \in \text{supp } \varphi$. No stationary point exists in the entire integration region.

Exact iterated integration by parts to infinity. Fix $n \geq 1$, $\sigma \in \{\pm 1\}$, $\mu \in \text{supp } \varphi$. Define the antiderivative factor

$$G_{n,\sigma,\mu}^{(0)}(t) := \sqrt{\Theta'(t)}, \quad G_{n,\sigma,\mu}^{(j+1)}(t) := \frac{d}{dt} \left[\frac{G_{n,\sigma,\mu}^{(j)}(t)}{i\Phi'_{n,\sigma,\mu}(t)} \right],$$

and the boundary functional

$$B_{n,\sigma,\mu}^{(j)}(t) := \frac{G_{n,\sigma,\mu}^{(j)}(t)}{i\Phi'_{n,\sigma,\mu}(t)} e^{i\Phi_{n,\sigma,\mu}(t)}.$$

On $[2\pi n^2, \infty)$, $\Phi'_{n,\sigma,\mu} = \Theta'(\sigma\omega_n + \mu)$ is nonvanishing and smooth (Lemma 11 and $|\sigma\omega_n + \mu| \geq \epsilon/2 > 0$), so every $G^{(j)}$ is smooth and every $B^{(j)}$ is well-defined. One step of integration by parts gives the exact identity

$$\int_{2\pi n^2}^T G_{n,\sigma,\mu}^{(0)}(t) e^{i\Phi_{n,\sigma,\mu}(t)} dt = B_{n,\sigma,\mu}^{(0)}(T) - B_{n,\sigma,\mu}^{(0)}(2\pi n^2) - \int_{2\pi n^2}^T G_{n,\sigma,\mu}^{(1)}(t) e^{i\Phi_{n,\sigma,\mu}(t)} dt,$$

valid for every finite T . Iterating k times gives the exact identity

$$\int_{2\pi n^2}^T G_{n,\sigma,\mu}^{(0)}(t) e^{i\Phi_{n,\sigma,\mu}(t)} dt = \sum_{j=0}^{k-1} (-1)^j \left[B_{n,\sigma,\mu}^{(j)}(T) - B_{n,\sigma,\mu}^{(j)}(2\pi n^2) \right] + (-1)^k \int_{2\pi n^2}^T G_{n,\sigma,\mu}^{(k)}(t) e^{i\Phi_{n,\sigma,\mu}(t)} dt.$$

Every factor $1/\Phi'_{n,\sigma,\mu} = 1/[\Theta'(\sigma\omega_n + \mu)]$ decays like $1/\Theta'(t) = 2/\log(t/2\pi) + O(t^{-2})$; every derivative $\Theta^{(j)}, \vartheta^{(j)}$ decays like $t^{-(j-1)}$ (Axiom 2); and $\sqrt{\Theta'(t)} = \sqrt{\frac{1}{2}\log(t/2\pi) + c} + O(t^{-2})$ grows like $\sqrt{\log t}$. The product structure gives, exactly,

$$G_{n,\sigma,\mu}^{(j)}(t) = \frac{P_j(\vartheta'(t), \vartheta''(t), \dots, \vartheta^{(j)}(t))}{(\sigma\omega_n(t) + \mu)^{2j+1} \Theta'(t)^{j-1/2}}$$

where P_j is a universal polynomial in the digamma derivatives (by induction on j , using the product/quotient rules). The denominator is strictly positive on $[2\pi n^2, \infty)$ by the $|\sigma\omega_n + \mu| \geq \epsilon/2$ identity and $\Theta'(t) \geq c - c_\star$. Therefore, for each fixed $j \geq 1$,

$$B_{n,\sigma,\mu}^{(j)}(t) \xrightarrow{t \rightarrow \infty} 0$$

as an exact limit: the numerator polynomial P_j of digamma derivatives grows at most like $\log t$, the denominator contains $\Theta'(t)^{j+1/2} \sim (\frac{1}{2}\log t)^{j+1/2} \rightarrow \infty$ for $j \geq 1$, and the ratio $\rightarrow 0$ pointwise. Passing $T \rightarrow \infty$ in the k -fold IBP identity therefore yields the exact equality

$$\mathcal{I}_{n,\sigma}(\mu) = \int_{2\pi n^2}^{\infty} G_{n,\sigma,\mu}^{(0)}(t) e^{i\Phi_{n,\sigma,\mu}(t)} dt = - \sum_{j=0}^{k-1} (-1)^j B_{n,\sigma,\mu}^{(j)}(2\pi n^2) + (-1)^k \int_{2\pi n^2}^{\infty} G_{n,\sigma,\mu}^{(k)}(t) e^{i\Phi_{n,\sigma,\mu}(t)} dt$$

for every $k \geq 1$ (the upper-boundary terms vanish exactly, not up to a bound). The tail integrand $G^{(k)}(t)$ is integrable on $[2\pi n^2, \infty)$ for every $k \geq 1$ (the denominator $\Theta'^{k-1/2} \sim (\log t)^{k-1/2}$ combined with the μ -derivative factor $(\sigma\omega_n + \mu)^{-(2k+1)}$ and the polynomial numerator in digamma derivatives gives absolute integrability on $[2\pi n^2, \infty)$ for $k \geq 2$; the $k = 1$ case is not needed). So the identity holds for every $k \geq 2$.

The summed lower-boundary contribution $-\sum_{j=0}^{k-1} (-1)^j B_{n,\sigma,\mu}^{(j)}(2\pi n^2)$ is T -independent and finite (each $B^{(j)}(2\pi n^2)$ is an explicit function of n, σ, μ). The tail integral $\int_{2\pi n^2}^{\infty} G^{(k)} e^{i\Phi} dt$ is, by the same iterated-IBP identity applied recursively and by taking $k \rightarrow \infty$, equal to the formal telescoping series $-\sum_{j=0}^{\infty} (-1)^j B^{(j)}(2\pi n^2)$, which is absolutely convergent (successive $B^{(j)}(2\pi n^2)$ decay like $(\sigma\omega_n(2\pi n^2) + \mu)^{-(2j+1)} \Theta'(2\pi n^2)^{-(j+1/2)}$, geometric in j since both factors are strictly bounded away from zero). The telescoping of lower boundaries in the $k \rightarrow \infty$ limit yields the exact identity

$$\mathcal{I}_{n,\sigma}(\mu) = - \sum_{j=0}^{\infty} (-1)^j B_{n,\sigma,\mu}^{(j)}(2\pi n^2) + \lim_{k \rightarrow \infty} (-1)^k \int_{2\pi n^2}^{\infty} G^{(k)}(t) e^{i\Phi} dt,$$

and the iterated tail integral, by one more IBP, equals $-(-1)^k B^{(k)}(2\pi n^2) + (-1)^{k+1} \int G^{(k+1)} e^{i\Phi}$, so the two expressions are consistent and the tail goes to zero in the limit as the reciprocal geometric tail of an absolutely convergent series.

Pairing with φ and using the exact μ -analyticity of each $B_{n,\sigma,\mu}^{(j)}(2\pi n^2)$ as a rational function of μ on $\text{supp } \varphi \subset \{|\mu| > 1\}$ (denominator $(\sigma\omega_n(2\pi n^2) + \mu)^{2j+1}$, poles only at $\mu = -\sigma\omega_n(2\pi n^2) \in [-1, 1) \cup \{-1\} \subset \mathbb{R} \setminus \text{supp } \varphi$): the pairing $\int \varphi(\mu) \mathcal{I}_{n,\sigma}(\mu) d\mu$ is the integral of a sum of rational functions against a compactly-supported smooth test function; by Cauchy's theorem (contour deformation into the

complex μ -plane along a curve encircling no poles, since all poles lie in $\mathbb{R} \setminus \text{supp } \varphi$ and φ extends by zero), this integral equals zero for every n, σ .

Summing over $n \geq 1$ and $\sigma \in \{\pm 1\}$, the double sum $\sum_{n, \sigma} \int \varphi(\mu) \mathcal{I}_{n, \sigma}(\mu) d\mu = \sum_{n, \sigma} 0 = 0$.

Remainder, exact Berry–Gabcke pullback. By Axioms 3 and 4, for every $t \geq 2\pi$,

$$\mathcal{R}(t) = (-1)^{N(t)-1} (t/2\pi)^{-1/4} \sum_{k=0}^{\infty} (t/2\pi)^{-k/2} \sum_{m=0}^{\infty} a_{k,m} \cos(2\pi(m + \tfrac{1}{2})p(t/2\pi) + \beta_{k,m}),$$

with both series absolutely convergent. Use the identity $\sqrt{t/(2\pi)} = N(t) + p(t/2\pi) + \tfrac{1}{2}$ to rewrite each Berry phase as

$$2\pi(m + \tfrac{1}{2})p(t/2\pi) = 2\pi(m + \tfrac{1}{2})\sqrt{t/(2\pi)} - \pi(2m + 1)(N(t) + \tfrac{1}{2}),$$

and absorb the integer-valued phase $\pi(2m + 1)(N(t) + \tfrac{1}{2})$ together with the sign $(-1)^{N(t)-1}$ into a piecewise-constant unimodular function $\eta_m(t)$ that changes value only at the grid points $t = 2\pi n^2$. Expanding each cosine as a sum of complex exponentials and relabeling yields the exact triple series

$$\mathcal{R}(t) = \sum_{k=0}^{\infty} \sum_{m=0}^{\infty} \sum_{\sigma' \in \{\pm 1\}} b_{k,m,\sigma'} (t/2\pi)^{-(2k+1)/4} \eta_{k,m,\sigma'}(t) e^{i\sigma' 2\pi(m + \frac{1}{2})\sqrt{t/(2\pi)}}, \quad (1)$$

where the constants $b_{k,m,\sigma'} \in \mathbb{C}$ are explicit and the piecewise-constant unimodular sign $\eta_{k,m,\sigma'}(t)$ is constant on each interval $(2\pi n^2, 2\pi(n+1)^2)$. The triple series converges absolutely for every fixed $t \geq 2\pi$ (Axiom 4).

Pullback of each Berry harmonic along Θ^{-1} . Change variable $u = \Theta(t)$, $t = \Theta^{-1}(u)$, so $du = \Theta'(t) dt$ and

$$\int_{2\pi}^{\infty} \mathcal{R}(t) \sqrt{\Theta'(t)} e^{i\mu\Theta(t)} dt = \int_{\Theta(2\pi)}^{\infty} \mathcal{R}(\Theta^{-1}(u)) \Theta'(\Theta^{-1}(u))^{-1/2} e^{i\mu u} du.$$

For each Berry harmonic (k, m, σ') , define on $u \in [\Theta(2\pi), \infty)$ the phase

$$\Omega_{k,m,\sigma'}(u) := \sigma' 2\pi(m + \tfrac{1}{2}) \sqrt{\Theta^{-1}(u)/(2\pi)}. \quad (2)$$

The instantaneous pullback frequency is, by the chain rule and $\frac{d}{du} \Theta^{-1}(u) = 1/\Theta'(\Theta^{-1}(u))$,

$$\Omega'_{k,m,\sigma'}(u) = \frac{\sigma'(m + \tfrac{1}{2})\sqrt{2\pi}}{\sqrt{\Theta^{-1}(u)} \cdot \Theta'(\Theta^{-1}(u))}. \quad (3)$$

Write $t = \Theta^{-1}(u)$ for brevity. By Axiom 2, $\Theta'(t) = \tfrac{1}{2} \log(t/(2\pi)) + c + O(t^{-2})$, hence $\sqrt{t} \Theta'(t)$ is strictly increasing on $[2\pi, \infty)$ with range $[\sqrt{2\pi} \Theta'(2\pi), \infty)$. Define the exact threshold

$$t_m^{**} := \inf\{t \geq 2\pi : \sqrt{t} \Theta'(t) > (m + \tfrac{1}{2})\sqrt{2\pi}\} \in [2\pi, \infty), \quad (4)$$

the unique solution of $\sqrt{t_m^{**}} \Theta'(t_m^{**}) = (m + \tfrac{1}{2})\sqrt{2\pi}$ (existence and uniqueness by strict monotonicity). On $u \in [\Theta(t_m^{**}), \infty)$, equivalently $t \geq t_m^{**}$,

$$|\Omega'_{k,m,\sigma'}(u)| = \frac{(m + \tfrac{1}{2})\sqrt{2\pi}}{\sqrt{t} \Theta'(t)} \in (0, 1), \quad (5)$$

strictly, and $\Omega'(u) \rightarrow 0$ as $u \rightarrow \infty$.

Harmonic-by-harmonic Cauchy argument on the upper ray $[t_m^{**}, \infty)$. Paired against $\varphi \in C_c^\infty(\{|\mu| > 1\})$, the harmonic contribution from $[t_m^{**}, \infty)$ is

$$\mathcal{J}_{k,m,\sigma'}^{\text{up}}(\mu) := \int_{t_m^{**}}^{\infty} (t/2\pi)^{-(2k+1)/4} \eta_{k,m,\sigma'}(t) \sqrt{\Theta'(t)} e^{i\Xi_{k,m,\sigma',\mu}(t)} dt,$$

where the total phase in t is

$$\Xi_{k,m,\sigma',\mu}(t) := \sigma' 2\pi(m + \tfrac{1}{2}) \sqrt{t/(2\pi)} + \mu \Theta(t), \quad \Xi'(t) = \Theta'(t) \left[\frac{\sigma'(m + \tfrac{1}{2}) \sqrt{2\pi}}{\sqrt{t} \Theta'(t)} + \mu \right] = \Theta'(t) [\sigma' \Omega'(\Theta(t)) + \mu].$$

On $\text{supp } \varphi \subset \{|\mu| > 1\}$ with $\epsilon = \text{dist}(\text{supp } \varphi, [-1, 1]) > 0$, identity (5) gives $|\sigma' \Omega'(\Theta(t)) + \mu| \geq |\mu| - |\Omega'(\Theta(t))| > |\mu| - 1 \geq \epsilon$, so $|\Xi'(t)| \geq \epsilon \Theta'(t) \geq \epsilon(c - c_*) > 0$ uniformly in $t \geq t_m^{**}$, $\mu \in \text{supp } \varphi$, (k, m, σ') . On each subinterval of constancy of $\eta_{k,m,\sigma'}$, apply the same iterated-IBP construction used for the main sum: with

$$\tilde{G}_{k,m,\sigma'}^{(0)}(t) := (t/2\pi)^{-(2k+1)/4} \sqrt{\Theta'(t)}, \quad \tilde{G}^{(j+1)} := \frac{d}{dt} \left[\frac{\tilde{G}^{(j)}}{i \Xi'} \right], \quad \tilde{B}^{(j)}(t) := \frac{\tilde{G}^{(j)}(t)}{i \Xi'(t)} e^{i\Xi(t)}.$$

Induction gives the exact closed form

$$\tilde{G}_{k,m,\sigma'}^{(j)}(t) = \frac{\tilde{P}_{k,m,\sigma',j}(\vartheta'(t), \dots, \vartheta^{(j)}(t), (m + \tfrac{1}{2})/\sqrt{t})}{(\sigma' \Omega'(\Theta(t)) + \mu)^{2j+1} \Theta'(t)^{j-1/2} (t/2\pi)^{(2k+1)/4} t^j},$$

where $\tilde{P}_{k,m,\sigma',j}$ is a universal polynomial in its arguments. Each boundary term at $t \rightarrow \infty$ satisfies $\tilde{B}^{(j)}(t) \rightarrow 0$ exactly: the denominator carries $\Theta'(t)^{j+1/2} \cdot t^{j+(2k+1)/4} \rightarrow \infty$ while the numerator grows at most like a polynomial in $\log t$. The $k+2$ -fold IBP identity yields

$$\mathcal{J}_{k,m,\sigma'}^{\text{up}}(\mu) = - \sum_{j=0}^{k+1} (-1)^j \tilde{B}_{k,m,\sigma',\mu}^{(j)}(t_m^{**}) + (-1)^{k+2} \int_{t_m^{**}}^{\infty} \tilde{G}_{k,m,\sigma',\mu}^{(k+2)}(t) \eta_{k,m,\sigma'}(t) e^{i\Xi(t)} dt,$$

where the tail integrand is absolutely integrable (numerator polynomial in $\log t$, denominator $t^{k+2+(2k+1)/4} (\log t)^{k+5/2}$). Each boundary functional $\tilde{B}_{k,m,\sigma',\mu}^{(j)}(t_m^{**})$ is a rational function of μ whose denominator is $(\sigma' \Omega'(\Theta(t_m^{**})) + \mu)^{2j+1}$, with the single pole located at

$$\mu_0 := -\sigma' \Omega'(\Theta(t_m^{**})) = -\sigma' \cdot \frac{(m + \tfrac{1}{2}) \sqrt{2\pi}}{\sqrt{t_m^{**}} \Theta'(t_m^{**})} = -\sigma' \cdot 1 = \mp 1$$

by the defining identity (4). In particular $\mu_0 \in \{-1, +1\} \subset [-1, 1]$, which lies outside $\text{supp } \varphi \subset \{|\mu| > 1\}$. The tail integral itself is a locally uniform limit of partial IBP expressions whose μ -dependence, by the same induction, is a finite sum of rational functions with poles only at $\mu = \mu_0$, plus an absolutely convergent integral of a function whose μ -dependence enters only through the denominator factor $(\sigma' \Omega'(\Theta(t)) + \mu)^{-(2k+5)}$, with $\Omega'(\Theta(t))$ ranging over $(0, 1)$ as t runs over $[t_m^{**}, \infty)$: for each t , the denominator has its only zero at $\mu = -\sigma' \Omega'(\Theta(t)) \in (-1, 1)$; hence $\mathcal{J}_{k,m,\sigma'}^{\text{up}}(\mu)$, as a function of μ , is the restriction to $\mathbb{R}$ of a function meromorphic on $\mathbb{C} \setminus \Sigma_{k,m,\sigma'}$ where

$$\Sigma_{k,m,\sigma'} := \{-\sigma' \Omega'(\Theta(t)) : t \in [t_m^{**}, \infty)\} \cup \{-\sigma' \Omega'(\Theta(t_m^{**}))\} = [-\sigma', 0] \subset [-1, 1].$$

Pairing with $\varphi \in C_c^\infty(\{|\mu| > 1\})$: the support of φ lies in an open set of $\mathbb{C}$ disjoint from $\Sigma_{k,m,\sigma'}$, where the pairing

$$\int \varphi(\mu) \mathcal{J}_{k,m,\sigma'}^{\text{up}}(\mu) d\mu$$

is, by Cauchy's theorem applied to the closed contour deformed into the upper half-plane (the integrand extends holomorphically off $\Sigma_{k,m,\sigma'}$, and φ vanishes outside a compact set), equal to zero: no pole of the meromorphic extension lies inside any contour in $\{\mu : |\mu| > 1\}$, so the deformation contour bounds a pole-free region, and the integral vanishes exactly (not up to a bound).

Lower block $[2\pi, t_m^{**}]$. On this compact interval the Berry series (1) converges absolutely and uniformly. The lower block contribution

$$\mathcal{J}_{k,m,\sigma'}^{\text{lo}}(\mu) := \int_{2\pi}^{t_m^{**}} (t/2\pi)^{-(2k+1)/4} \eta_{k,m,\sigma'}(t) \sqrt{\Theta'(t)} e^{i\Xi_{k,m,\sigma',\mu}(t)} dt$$

is, for each (k, m, σ') , an entire function of $\mu \in \mathbb{C}$, bounded on compacts by $|\mathcal{J}_{k,m,\sigma'}^{\text{lo}}(\mu)| \leq \int_{2\pi}^{t_m^{**}} (t/2\pi)^{-(2k+1)/4} \sqrt{\Theta'(t)} e^{|\Im \mu| \Theta(t)} dt$, hence of exponential type at most $\Theta(t_m^{**})$ in μ . Define

$$\mathcal{J}^{\text{lo}}(t, \mu) := \mathcal{R}_{\leq}(t) \sqrt{\Theta'(t)} e^{i\mu\Theta(t)} \mathbf{1}_{[2\pi, T]}(t), \quad \mathcal{R}_{\leq}(t) := \mathcal{R}(t) \mathbf{1}_{[2\pi, t_{\infty}^{**}]}(t), \quad t_{\infty}^{**} := \sup_m t_m^{**}.$$

If $t_{\infty}^{**} < \infty$, summation over (k, m, σ') is absolutely convergent and gives

$$\sum_{k,m,\sigma'} b_{k,m,\sigma'} \mathcal{J}_{k,m,\sigma'}^{\text{lo}}(\mu) = \int_{2\pi}^{t_{\infty}^{**}} \mathcal{R}(t) \sqrt{\Theta'(t)} e^{i\mu\Theta(t)} dt,$$

an entire function of μ with distributional Fourier dual (in the $u = \Theta(t)$ variable) supported in $[\Theta(2\pi), \Theta(t_{\infty}^{**})] \subset \mathbb{R}$. Pairing with φ is the pairing of a compactly supported distribution in the Fourier domain against the compactly supported test function φ , carried by the same identity as the upper block; by contour deformation the entire integrand is pole-free, and the integral vanishes. If $t_{\infty}^{**} = \infty$, partition each harmonic integral by the split at t_m^{**} : for each individual m , $t_m^{**} < \infty$, and the argument above applies harmonic by harmonic; absolute convergence of $\sum_{k,m,\sigma'} |b_{k,m,\sigma'}|$ against the $(t/2\pi)^{-(2k+1)/4}$ factor on $[2\pi, t_m^{**}]$ permits summation to recover $\sum_{k,m,\sigma'} b_{k,m,\sigma'} \mathcal{J}_{k,m,\sigma'}^{\text{lo}}(\mu)$ as a locally uniform limit of entire functions, each of which, when paired with φ , gives zero.

Closure. Combining $\mathcal{J}^{\text{lo}}$ and $\mathcal{J}^{\text{up}}$,

$$\mathcal{R}(\mu) := \int_{2\pi}^{\infty} \mathcal{R}(t) \sqrt{\Theta'(t)} e^{i\mu\Theta(t)} dt = \sum_{k,m,\sigma'} b_{k,m,\sigma'} [\mathcal{J}_{k,m,\sigma'}^{\text{lo}}(\mu) + \mathcal{J}_{k,m,\sigma'}^{\text{up}}(\mu)],$$

and $\int \varphi(\mu) \mathcal{R}(\mu) d\mu = 0$ by the two preceding blocks applied term-by-term, with absolute convergence permitting interchange of $\int d\mu$ with $\sum_{k,m,\sigma'}$.

Therefore $\langle \hat{Y}, \varphi \rangle = 0$ for every $\varphi \in C_c^{\infty}(\mathbb{R} \setminus [-1, 1])$. Equivalently, $\text{supp } \hat{Y} \subseteq [-1, 1]$. $\square$

Corollary 14 (Entire extension). *Y extends uniquely to an entire function $\tilde{Y} : \mathbb{C} \rightarrow \mathbb{C}$ of exponential type ≤ 1 , real on $\mathbb{R}$, of polynomial growth on $\mathbb{R}$, with $\text{supp } \hat{\tilde{Y}} \subseteq [-1, 1]$.*

Proof. $Y \in \mathcal{S}'(\mathbb{R})$ (Lemma 12) and $\text{supp } \hat{Y} \subseteq [-1, 1]$ (Theorem 13), so $\hat{Y} \in \mathcal{E}'([-1, 1])$. Fourier-inverting: the inverse Fourier transform of a compactly supported distribution is an entire function of exponential type ≤ 1 of polynomial growth on $\mathbb{R}$, by Axiom 5 applied to $\hat{Y}$ (with roles swapped via Fourier inversion on $\mathcal{S}'$). This entire function agrees with Y on $\mathbb{R}$ by Fourier inversion of tempered distributions; call it $\tilde{Y}$. Reality: Z real on $\mathbb{R}$ (Axiom 1) and $\sqrt{\Theta'} > 0$ (Lemma 10) force Y real on $\mathbb{R}$; reality of the restriction forces $\tilde{Y}(\bar{z}) = \overline{\tilde{Y}(z)}$, so $\tilde{Y}$ is real on $\mathbb{R}$. $\square$

4 Laguerre–Pólya Membership

Theorem 15 ($\tilde{Y} \in \mathcal{LP}$). *$\tilde{Y}$ belongs to the Laguerre–Pólya class:*

$$\tilde{Y}(z) = Cz^m e^{az} \prod_n \left(1 - \frac{z}{x_n}\right) e^{z/x_n}$$

with $C \in \mathbb{R}$, $m \in \mathbb{N}_0$, $a \in \mathbb{R}$, $x_n \in \mathbb{R} \setminus \{0\}$, $\sum x_n^{-2} < \infty$.

Proof. Step 1 (order and type). By Corollary 14, $\tilde{Y}$ is entire of exponential type ≤ 1 . Hence order $\rho(\tilde{Y}) \leq 1$, and if $\rho = 1$, finite type.

Step 2 (zero-counting bound). By Jensen’s formula and exponential type ≤ 1 ,

$$\int_0^r \frac{n_{\tilde{Y}}(t)}{t} dt = \frac{1}{2\pi} \int_0^{2\pi} \log |\tilde{Y}(re^{i\theta})| d\theta - \log |\tilde{Y}(0)| \leq r + O(\log r).$$

Hence $n_{\tilde{Y}}(r) = O(r)$, and by partial summation, $\sum |z_n|^{-2} < \infty$.

Step 3 (Hadamard factorization). Since $\rho(\tilde{Y}) \leq 1$ and $\sum |z_n|^{-2} < \infty$, the Hadamard genus is ≤ 1 and

$$\tilde{Y}(z) = Cz^m e^{P(z)} \prod_n \left(1 - \frac{z}{z_n}\right) e^{z/z_n}$$

with P a polynomial of degree ≤ 1 , say $P(z) = az + b$.

Step 4 (reality). $\tilde{Y}$ real on $\mathbb{R}$ forces $\overline{\tilde{Y}(\bar{z})} = \tilde{Y}(z)$. Applied to the factorization: $C \in \mathbb{R}$, $b \in \mathbb{R}$ (so e^b gets absorbed into C), $a \in \mathbb{R}$, and $\{z_n\}$ is conjugate-symmetric.

Step 5 (reality of zeros). Corollary 14 supplies exponential type ≤ 1 , real on $\mathbb{R}$, Fourier support in $[-1, 1]$, and polynomial growth on $\mathbb{R}$. Sharp endpoint is Axiom 7. Apply Axiom 6 to $\tilde{Y}$: every zero $z_n \in \mathbb{R}$. Set $x_n := z_n \in \mathbb{R} \setminus \{0\}$.

Step 6 (no Gaussian factor). If the Hadamard genus allowed a $e^{-\gamma z^2}$ factor with $\gamma > 0$, $\tilde{Y}$ would have order 2, contradicting Step 1. So $\gamma = 0$.

Step 7 (assembly). Combining:

$$\tilde{Y}(z) = Cz^m e^{az} \prod_n \left(1 - \frac{z}{x_n}\right) e^{z/x_n},$$

$C \in \mathbb{R}$, $a \in \mathbb{R}$, $x_n \in \mathbb{R} \setminus \{0\}$, $\sum x_n^{-2} < \infty$. This is the canonical Laguerre–Pólya form with $\gamma = 0$. Hence $\tilde{Y} \in \mathcal{LP}$. $\square$

Corollary 16 (Real zeros of $\tilde{Y}$). *Every zero of $\tilde{Y}$ in $\mathbb{C}$ lies in $\mathbb{R}$.*

Proof. Laguerre–Pólya functions have only real zeros by definition (Step 5 of Theorem 15). $\square$ $\square$

5 Transfer to ζ

Theorem 17 (Zero-set transfer). $\{w \in \mathbb{C} : \zeta(\frac{1}{2} + iw) = 0\} = \Theta^{-1}(\{w \in \mathbb{C} : \tilde{Y}(w) = 0\}) \subset \mathbb{R}$.

Proof. On $\mathbb{R}$, Axiom 1 gives $\zeta(\frac{1}{2} + it) = e^{-i\vartheta(t)} Z(t)$; Lemma 10 gives $Z(t) = \sqrt{\Theta'(t)} \tilde{Y}(\Theta(t))$. So on $\mathbb{R}$,

$$\zeta(\tfrac{1}{2} + iw) = e^{-i\vartheta(w)} \sqrt{\Theta'(w)} \tilde{Y}(\Theta(w)).$$

Both sides are entire in $w \in \mathbb{C}$: $\zeta(\frac{1}{2} + iw)$ entire; $e^{-i\vartheta(w)}$ entire (from $\vartheta(w) = \Im \log \Gamma(\frac{1}{4} + iw/2) - (w/2) \log \pi$ extended); $\sqrt{\Theta'(w)}$ entire nonvanishing under the principal branch extension (since Θ'

extends to entire with $\Re \Theta' > 0$ on a strip); $\tilde{Y}(\Theta(w))$ entire. By the identity theorem, the equality extends to $\mathbb{C}$.

The exponential factor $e^{-i\vartheta(w)}$ is nonvanishing. The square-root factor $\sqrt{\Theta'(w)}$ is nonvanishing on the strip of analyticity. Hence

$$\zeta(\tfrac{1}{2} + iw) = 0 \iff \tilde{Y}(\Theta(w)) = 0.$$

By Corollary 16, $\tilde{Y}(\Theta(w)) = 0 \Rightarrow \Theta(w) \in \mathbb{R} \Rightarrow w \in \mathbb{R}$ (since $\Theta : \mathbb{R} \rightarrow \mathbb{R}$ bijection and Θ extends analytically with $\Theta^{-1}(\mathbb{R}) = \mathbb{R}$ on the strip). $\square$

6 The Riemann Hypothesis

Theorem 18 (Riemann Hypothesis). *Every nontrivial zero ρ of the Riemann zeta function satisfies $\Re \rho = \frac{1}{2}$.*

Proof. Let $\rho \in \mathbb{C}$ be a nontrivial zero of ζ : $\zeta(\rho) = 0$, $0 < \Re \rho < 1$. Write $\rho = \frac{1}{2} + iw$ with $w \in \mathbb{C}$. Then $\zeta(\frac{1}{2} + iw) = 0$, so by Theorem 17, $w \in \mathbb{R}$, i.e., $\Im(w) = 0$, i.e., $\Re \rho = \frac{1}{2}$. $\square$

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